

HANDOUT TITLE

SPRING 2026 · COURSE NAME

1. Energy and Conservation

A particle of mass m moves in a one-dimensional potential $V(x)$. The total mechanical energy is

$$E = \frac{1}{2}mv^2 + V(x), \quad (1)$$

which is conserved when no external forces act.

Exercise 1.1

Consider a particle of mass m in the potential $V(x) = \frac{1}{2}kx^2$.

- a) Write down the equation of motion using Newton's second law.
- b) Show that $E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$ is conserved along solutions.

The result can be verified by differentiating E with respect to time and using the equation of motion from part (a).

- c) Find the angular frequency of small oscillations.

Exercise 1.2

A block of mass m slides down a frictionless inclined plane that makes an angle θ with the horizontal.

- a) Find the acceleration along the plane.
- b) How long does it take to slide a distance L from rest?
- c) What is the speed at the bottom?

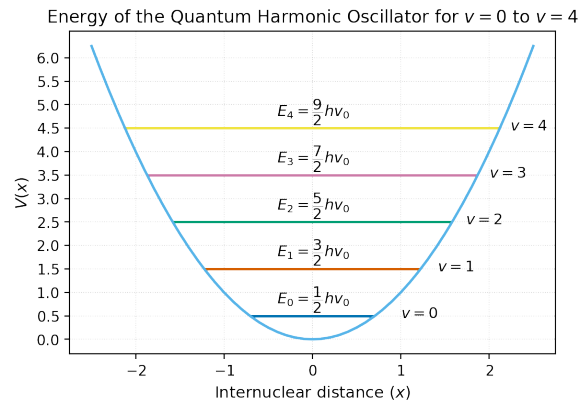


Figure 1: Schematic figure for the topic.

2. Circular Motion

In uniform circular motion, the centripetal acceleration has magnitude $a_c = v^2/r$, directed toward the center of the circle.

Exercise 2.1

A satellite moves in a circular orbit at radius r around a planet of mass M .

- Show that the orbital speed is $v = \sqrt{GM/r}$, where G is Newton's gravitational constant.
- Express the orbital period T in terms of r , M , and G .